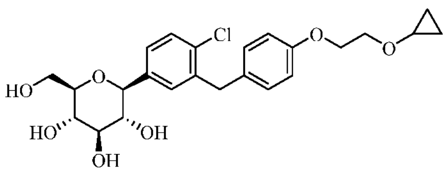
glucosuria by increasing the urinary excretion of glucose. SGLT2 inhibitor indicated for the treatment of type 2 diabetes mellitus, heart failure, and chronic kidney disease. Bexagliflozin is an inhibitor of sodium-glucose co-transporter 2 (SGLT2), the compound is investigated in lowering hemoglobin Ale (HbAlc) levels in patients with type 2 diabetes mellitus (T2DM) and moderate renal impairment. The compound is for therapeutic intervention in diabetes and related disorders, SGLT2 is localized in the renal proximal tubule and is reportedly responsible for the majority of glucose reuptake by the kidneys and is marketed under the proprietary name Brenzavvy, Bexacat by THERAXOSBIO LLC is chemically named as (2S,3R,4R,5S,6R)-2-(4-Chloro-3-(4-(2-cyclopropoxyethoxy)benzyl)phenyl)-6-(hydroxymethyl)tetrahydro-2H-pyran-3,4,5-triol and has the following chemical structure



Formula I

[0005] Bexagliflozin shows pharmaceutical activity by functioning as a Sodium-glucose co-transporter-2 (SGLT-2) inhibitor and thus is indicated for the treatment of type 2 diabetes mellitus.

[0006] Several synthetic methods have been reported in the literature to prepare Bexagliflozin, (2S,3R,4R,5S,6R)-2-(4-Chloro-3-(4-(2-cyclopropoxyethoxy)benzyl)phenyl)-6-(hydroxymethyl)tetrahydro-2H-pyran-3,4,5-triol and its intermediates.

[0007] U.S. Patent No.7838499 discloses the below process to prepare compound of formula (I) as per the following synthetic scheme